3. A biased spinner can land on the numbers 1, 2, 3, 4 or 5 with the following probabilities.

Number on spinner	1	2	3	4	5
Probability	0.3	0.1	0.2	0.1	0.3

The spinner will be spun 80 times and the mean of the numbers it lands on will be calculated.
Find an estimate of the probability that this mean will be greater than 3.25

(6)

7. A six-sided die has sides labelled 1, 2, 3, 4, 5 and 6

The random variable S represents the score when the die is rolled.

Alicia rolls the die 45 times and the mean score, $\overline{S}$, is calculated.

Assuming the die is fair and using a suitable approximation,

(a) find, to 3 significant figures, the value of k such that $P(\overline{S} < k) = 0.05$

(8)

(b) Explain the relevance of the Central Limit Theorem in part (a).

(2)

Alicia considers the following hypotheses:

H_0: The die is fair

H_1: The die is not fair

If $\overline{S} < 3.1$ or $\overline{S} > 3.9$, then H_0 will be rejected.

Given that the true distribution of S has mean 4 and variance 3

(c) find the power of this test.

(3)

(d) Describe what would happen to the power of this test if Alicia were to increase the number of rolls of the die.
Give a reason for your answer.

(2)

3. A courier delivers parcels. The random variable X represents the number of parcels delivered successfully each day by the courier where $X \sim B(400, 0.64)$

A random sample $X_1, X_2, \ldots X_{100}$ is taken.

Estimate the probability that the mean number of parcels delivered each day by the courier is greater than 257

(4)

5. A random sample of 150 observations is taken from a geometric distribution with parameter 0.3

Estimate the probability that the mean of the sample is less than 3.45

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

4. There are 32 students in a class.

Each student rolls a fair die repeatedly, stopping when their total number of sixes is 4
Each student records the total number of times they rolled the die.

Estimate the probability that the mean number of rolls for the class is less than 27.2

(6)

4. Every morning Geethaka repeatedly rolls a fair, six-sided die until he rolls a 3 and then he stops. The random variable X represents the number of times he rolls the die each morning.

(a) Suggest a suitable model for the random variable X

(1)

(b) Show that $P(X \leqslant 3) = \dfrac{91}{216}$

(2)

After 64 mornings Geethaka will calculate the mean number of times he rolled the die.

(c) Estimate the probability that the mean number of rolls is between 5.6 and 7.2

(5)

Nira wants to check Geethaka's die to decide whether or not the probability of rolling a 3 with his die is less than $\dfrac{1}{6}$

Nira rolls the die repeatedly until she rolls a 3
She obtains $x = 16$

(d) By carrying out a suitable test, determine what Nira's conclusion should be. You should state your hypotheses clearly and use a 5% level of significance.

(4)

4. A random sample of 100 observations is taken from a Poisson distribution with mean 2.3

Estimate the probability that the mean of the sample is greater than 2.5

(4)